BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Calculus II (MAT211)

Limit and Continuity

Limit, Continuity, Algebra of Limits

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Conducted by

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Definition of Limit

Limit

Let $U \subset \mathbb{R}^n$ be an open subset containing a neighborhood of $a \in \mathbb{R}^n$, except perhaps for the point a itself. Suppose $f : U \to \mathbb{R}^m$. We say that

$$\lim_{x \to a} f(x) = \ell$$

($f(x)$ approaches $\ell \in \mathbb{R}^m$ as x approaches a) if for every $\varepsilon > 0$ there is $\delta > 0$ so that

$$\|f(x) - \ell\| < \varepsilon \qquad \text{whenever} \qquad 0 < \|x - a\| < \delta.$$

Component wise Limit

Proposition

$$\lim_{x \to a} f(x) = \ell$$

if and only if

$$\lim_{x \to a} f_j(x) = \ell_j$$

for all $j = 1, \ldots, m$.

Examples of Limits

Problem

Fix a non-zero vector $b \in \mathbb{R}^n$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = b \cdot x$. Show that

$$\lim_{x \to a} f(x) = b \cdot a$$

$$b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \qquad f\left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}\right) = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

Given $\varepsilon > 0$,

Cauchy schwarz inequ.

$$\underbrace{|f(x) - b \cdot a|}_{<\varepsilon} = |b \cdot x - b \cdot a| = |b \cdot (x-a)| \leq \|b\| \cdot \|x-a\| < \cancel{\|b\|} \frac{\varepsilon}{\cancel{\|b\|}} < \varepsilon$$

$$\underbrace{\delta := \frac{\varepsilon}{\|b\|}}_{\text{set}}$$

$$\|x-a\| < \delta = ? \Rightarrow \|x-a\| < \frac{\varepsilon}{\|b\|}$$

Hence, $\forall \varepsilon > 0$ $\exists$ $\delta = \frac{\varepsilon}{\|b\|}$ then

$|f(x) - b \cdot a| < \varepsilon$ whenever $\|x-a\| < \delta$.

Examples of Limits

Problem

Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(\vec{x}) = \|\vec{x}\|^2$. Show that

$$\lim_{x \to a} f(x) = \|a\|^2$$

$$f\left(\underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{\vec{x}}\right) = \|\vec{x}\|^2 = \underbrace{x_1^2 + \cdots + x_n^2}_{\omega}$$

either $\vec{a} = \vec{0}$ or $\vec{a} \neq \vec{0}$

<u>case 1</u> $\vec{a} = \vec{0}$ $\qquad \lim_{x \to \vec{0}} f(x) = \vec{0}$

$1 \geq \varepsilon > 0$ $\qquad \delta = ?$ $\qquad \|x - 0\| < \delta \Rightarrow \|x\| < \delta = \varepsilon$

$$\|f(x) - \vec{0}\|^2 = \underbrace{\|\vec{x}\|^2}_{<\varepsilon} = \varepsilon^2 < \varepsilon \qquad \delta = \varepsilon$$

$\varepsilon > 1$ $\qquad \delta := \min\{1, \varepsilon\}$ $\qquad \|x\| < 1$

$$\|f(x) - 0\|^2 = \|x\|^2 < \|x\| < 1 < \varepsilon$$

<u>case-2</u> $\vec{a} \neq \vec{0}$

$$(x+a)\cdot(x-a) = x\cdot x - \cancel{x\cdot a} + \cancel{a\cdot x} - a\cdot a = x\cdot x - a\cdot a$$

Given $\varepsilon > 0$. $\qquad \|x - a\| < \delta = ?$

$$\left|f(x) - \|a\|^2\right| = |x\cdot x - a\cdot a| = |(x+a)\cdot(x-a)| \leq \|x+a\| \cdot \|x-a\|$$

$$< 3\|a\| \cdot \|x-a\| < 3\cancel{\|a\|} \cdot \frac{\varepsilon}{3\cancel{\|a\|}} < \varepsilon.$$

$$\delta := \frac{\varepsilon}{3\|a\|}$$

$\underbrace{\|x+a\|} \le \|x\| + \underbrace{\|a\|} < 2\|a\| + \|a\| = \underbrace{3\|a\|}$

Suppose $\boxed{\delta = \|a\|}$

$\|x\| = \|(x-a)+a\| \le \underbrace{\|x-a\|}_{\delta} + \|a\| = \delta + \|a\| < \|a\| + \|a\|$
$= 2\|a\|$

$\underbrace{\|x\| < 2\|a\|}$

then, $\delta := \min\left\{\|a\|, \dfrac{\varepsilon}{3\|a\|}\right\}$

<u>proof strategy</u>: $\underbrace{\text{start } \|x-a\| < \delta \text{ (set) then}}$

given $\varepsilon > 0$ $\quad \underbrace{\|f(x) - f(a)\| < \varepsilon}$

<u>working strategy</u> $\quad \varepsilon > 0 \quad \underbrace{\|f(x) - f(a)\| < \varepsilon}$

$\delta = ?$

$\varepsilon > 0$, $\delta =$ set

Examples of Limits

Problem

Let $f : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$ be defined by $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \boxed{\frac{x^2 y}{x^2+y^2}}$. Does $\lim_{x \to 0} f(x)$ exist?

$\mathbb{R}$

$|a| < \|\vec{v}\|, \ |b| < \|\vec{v}\|$

$\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ $\quad |x|^2 < \|\vec{x}\|^2, \ |y| < \|\vec{x}\|$

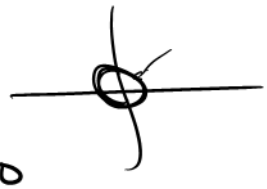
$x \le |x|$

$y \le |y| < \|\vec{x}\|$

$\vec{x}^2 + \vec{y}^2 = \|\vec{x}\|^2$

$$f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \frac{x^2 y}{x^2+y^2} < \frac{\|\vec{x}\|^2 \cdot \|\vec{x}\|}{\|\vec{x}\|^2} = \|\vec{x}\| \to 0$$

$\|\vec{x} - 0\| = \|\vec{x}\| < \delta = ?$



$\vec{x} \to 0, \ \|\vec{x}\| \to 0, \ f(\vec{x}) \to 0$

Given $\varepsilon > 0$

$\|\vec{x}\| = \delta = \varepsilon$

$\|x\| < \delta$

$\|f(x) - 0\| \le \|\vec{x}\| - 0| = \|\vec{x}\| < \varepsilon$

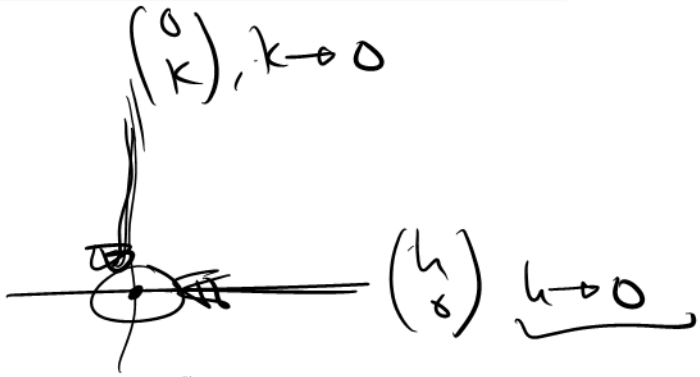
Examples of Limits

Problem

Let $f : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$ be defined by $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \frac{x^2}{x^2+y^2}$. Does $\lim_{x\to 0} f(x)$ exist?

$x^2 \le \|\vec{x}\|^2$

$f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) \le \frac{\cancel{\|\vec{x}\|^2}}{\cancel{\|\vec{x}\|^2}} = 1$



$$\lim_{h\to 0} f\begin{pmatrix} h \\ 0 \end{pmatrix} = \lim_{h\to 0} \frac{h^2}{h^2+0^2} = \lim_{h\to 0} \frac{\cancel{h^2}\,1}{\cancel{h^2}} = 1$$

$$\lim_{k\to 0} f\begin{pmatrix} 0 \\ k \end{pmatrix} = \lim_{k\to 0} \frac{0}{k^2} = 0$$

$1 < 1/2$

limit doesn't exist

Examples of Limits

Problem

Let $f : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$ be defined by $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \frac{xy}{x^4+y^2}$. Does $\lim_{x\to 0} f(x)$ exist?

Algebra of Limits

Theorem

Suppose f and g map a neighborhood of $a \in \mathbb{R}^n$ to $\mathbb{R}^m$ and k maps the same neighborhood to $\mathbb{R}$. Suppose $\lim_{x \to a} f(x) = \ell$, $\lim_{x \to a} g(x) = m$ and $\lim_{x \to a} k(x) = c$, then

- $\lim_{x \to a}(f(x) + g(x)) = \ell + m$,
- $\lim_{x \to a}(f(x) \cdot g(x)) = \ell \cdot m$,
- $\lim_{x \to a}(k(x)f(x)) = c\ell$.

Continuity

Definition

Let $U \subset \mathbb{R}^n$ be an open set containing a neighborhood of $a \in \mathbb{R}^n$, and let $f : U \to \mathbb{R}^m$. We say that f is continuous at a if

$$\lim_{x \to a} f(x) = f(a).$$

That is, f is continuous at a if for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$\|f(x) - f(a)\| < \varepsilon \qquad \text{whenever} \qquad \|x - a\| < \delta.$$

We say that f is continuous if it is continuous at every point in its domain.

Continuity and Open Sets

Proposition

Let $U \subset \mathbb{R}^n$ be an open set. The function $f : U \to \mathbb{R}^m$ is continuous at $a \in U$ if and only if for every open set $V \subset \mathbb{R}^m$, $f^{-1}(V)$ is open (the preimage of an open set is open).

Composition of Functions through open sets

Proposition

Suppose $U \subset \mathbb{R}^n$ and $W \subset \mathbb{R}^p$ are open, functions $f \circ g$ is defined (i.e., $g(x) \in U$ for all $x \in W$). $f : U \to \mathbb{R}^m$, $g : W \to \mathbb{R}^n$, and the composition of Then if f and g are continuous, so is $f \circ g$.

Now, Consider the first example.

Continuity and Sequences

Proposition

Suppose $U \subset \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^m$. Then f is continuous at a if and only if for every sequence $\{x_k\}$ of points in U converging to a the sequence $\{f(x_k)\}$ converges to $f(a)$.

Continuity and Level Sets

Proposition

Suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous. Then for any $c \in \mathbb{R}^m$, the level set

$$f^{-1}(\{c\}) = \{x \in \mathbb{R}^n : f(x) = c\}$$

is a closed set.

Thank You!

We'd love your questions and feedback.

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(Lectures, walkthroughs, and course updates)

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References

[1] THEODORE SHIFRIN, *MULTIVARIABLE MATHEMATICS: Linear Algebra, Multivariable Calculus, and Manifolds.*